

Vehicle Count Prediction - Dataset Overview

Context

This exercise focuses on Turkey as the primary evaluation country. In addition to Turkey, vehicle registration datasets from several European countries are provided, including: Ireland, Romania, Hungary, Spain, Poland, Slovakia, Belgium, Austria, Croatia, and others. The goal is to **predict missing vehicle counts**, `total_vehicles` , for certain combinations of brand, model, submodel, energy type and vehicle age. To ensure sufficient data coverage, only vehicle combinations that appear in at least seven countries have been retained.

Dataset Features

Each dataset contains the following variables:

Column	Description
<code>country_code</code>	Country identifier (e.g., TR for Turkey)
<code>year_stamp</code>	Observation year: 2024
<code>car_maker_name</code>	Vehicle brand / manufacturer
<code>car_segment_name</code>	Car segment (e.g., SUV, Sedan)
<code>car_type_name</code>	Specific car type
<code>market_segment</code>	Market segment / class
<code>body_style</code>	Body style (e.g., Hatchback, Sedan)
<code>energy</code>	Fuel type / energy source
<code>code_age</code>	Age of the model
<code>total_vehicles</code>	Target: total number of registered vehicles

Dataset information

country_name	car_maker_count	car_segment_count	car_type_count
Turkey	55	551	2362
Slovakia	70	781	3326
Belarus	59	574	2145
Croatia	68	721	3016

country_name	car_maker_count	car_segment_count	car_type_count
Spain	71	753	3103
Ireland	65	656	2647
Bosnia and Herzegovina	60	619	2479
Poland	70	756	3158
Romania	69	708	2965
Austria	73	781	3347
Hungary	71	790	3382

Dataset Split

- **Train dataset:** Contains all countries and most observations for Turkey.
- **Test dataset:** Contains **hidden rows** where `total_vehicles` is missing for **specific combinations** of: country x car_maker_name x car_type_name x car_segment_name x age model x energy

Important Note : All Land Rover for Turkey is to estimate it

- **Task:** Predict the `total_vehicles` values from the test dataset.

Evaluation Metric

Be careful **not to rely only on standard MSE or MAE**, because some brands or submodels have very few vehicles.

We recommend using **Weighted Mean Squared Error (WMSE)**:

$$\text{WMSE} = \frac{\sum_{i=1}^N w_i (y_i - \hat{y}_i)^2}{\sum_{i=1}^N w_i}$$

Where:

- y_i = true vehicle count
- $\hat{y}_i$ = predicted vehicle count
- $w_i = y_i$ (weight proportional, so high-volume brands dominate the loss)

Baseline

A heuristic baseline is provided to serve as a benchmark.

1. For each country, compute the total number of vehicles.

2. Convert raw vehicle counts into within-country shares:

share = vehicles for a given combination / total vehicles in the country

3. For each country and vehicle combination: Compute the average share observed for the same combination across all other countries.

4. Estimate the final vehicle count by multiplying the average share by the total vehicle volume of the target country.

The baseline predictions are available `EM_LYON_test_set_20260206.csv`, column `baseline_total_vehicles`

This baseline serves as a reference model. Your objective is to develop models that outperform this heuristic.

Notes for Students

- The problem should be treated as a regression task.
- Two datasets are available: Training dataset: `EM_LYON_train_set_20260206.csv` Test dataset: `EM_LYON_test_set_20260206.csv`